Financial Access and Consumption Smoothing

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Abstract

Does improving access to financial institutions facilitate consumption smoothing at different stages of development? Using data for 135 countries, we show that greater financial access raises consumption volatility relative income volatility in developing countries but lowers it in developed economies. We develop a parsimonious small-open economy model with household heterogeneity in financial access to show that the asymmetric nature of technological shocks experienced by developing and developed economies rationalizes this finding.

JEL Codes: E32, F41, O16

Keywords: Business cycles, Financial access, Small-open economy

1 Introduction

What is the impact of greater financial access on the volatility of consumption relative to income at different stages of development? The World Bank lists financial inclusion (individuals owning bank accounts) as central to achieving many of the Sustainable Development Goals.[1] Yet, the implications of financial access on macroeconomic volatility for countries at different stages of development, specifically consumption volatility, remain underexplored.

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[1] https://www.worldbank.org/en/topic/financialinclusion/overview

To examine this relationship empirically, we measure consumption-income volatility as the standard deviation of aggregate consumption relative to the standard deviation of aggregate income at business-cycle frequency between 1970 and 2019. Using data for 135 countries, we find that greater financial access correlates with higher consumption-income volatility in developing countries but with lower volatility in developed economies.

To explain this empirical pattern, we develop a small open-economy model that emphasizes the role of technological shocks – specifically, trend versus transitory shocks – in shaping the effect of financial access on consumption dynamics. The intuition is as follows: a positive trend shock generates a permanent increase in income growth, prompting financially included households to raise consumption by borrowing against higher expected future income. In contrast, a transitory income shock leads to a more muted consumption response, as households save part of the temporary gain. If developing economies predominantly experience trend shocks, then those with greater financial inclusion will exhibit higher consumption-income volatility. On the other hand, if developed economies mainly face transitory shocks, improvements in financial access facilitate consumption smoothing and reduce volatility.

The rest of the paper presents the key empirical result and the model that rationalizes it.

2 Empirical Fact

Financial inclusion and consumption-income volatility differ markedly between developing and developed countries: only 48% of individuals in developing countries own a bank account, compared to 89% in developed economies (Demirgüç-Kunt & Klapper, 2012), and consumption-income volatility is significantly higher in developing than developed nations (Aguiar & Gopinath, 2007; Uribe & Schmitt-Grohé, 2017). In this paper, we analyze the effect of financial access on consumption-income volatility *within* developing and developed economies.

We use annual time-series data on income and consumption in per capita real terms from the Penn World Table for the period 1970–2019 (Feenstra et al., 2015). We extract the cyclical component by applying the Hodrick-Prescott filter with a smoothing parameter of 6.25. Consumption-income volatility of a country is defined as the ratio of the standard deviation of log cyclical consumption to that of log cyclical income. Financial access is measured using the share of individuals in a country owning an account at any type of financial institution from the 2017 Global Findex Database (Demirgüç-Kunt & Klapper, 2012). We use the World

Bank classification to define high-income countries as developed and group all others (upper-middle, lower-middle and low income) as developing. The final sample includes 135 countries (Appendix Table A.1), comprising 43 developed and 92 developing economies.

We estimate a linear relationship between financial access and consumption-income volatility $\left(\frac{\sigma(ln(C))}{\sigma(ln(Y))}\right)$ for each development group, as specified in equation 1.

$$\left(\frac{\sigma(ln(C))}{\sigma(ln(Y))}\right)_i = \alpha + \beta \times \text{Financial Access}_i + \gamma X_i + \varepsilon_i \tag{1}$$

The specification controls for a range of factors (X_i) that may influence this relationship, including trade-openness, the cyclicality of government spending, terms-of-trade volatility, log population, exchange rate classification, political regime, dummies for debt, currency or banking crises and geographic region dummies. All regressions are weighted by the log of 2019 population to reduce the influence of small countries (see Appendix A.1 for details on data sources and controls used in the regressions, and Appendix Figure A.1 for summary statistics).

Table 1: Effect of financial access on consumption-income volatility

	Dependent variable: $\sigma(C)$ / $\sigma(Y)$	
	Developing	Developed
	(1)	(2)
	A: Full-sample	
Financial Access	0.009*	-0.015**
	(0.0048)	(0.0072)
Observations	92	43
R^2	0.266	0.590
	B: Removing outliers	
Financial Access	0.006*	-0.019***
	(0.0032)	(0.0060)
Observations	86	42
R^2	0.247	0.676

Note: Panel A includes the full-sample, whereas Panel B excludes countries with ratio of consumption volatility to income volatility of greater than equal to 3. All regression estimates are weighted by logarithm of country population in 2019. Robust standard errors are reported in parentheses. ***, ** and * represents the statistical significance at 1%, 5% and 10% levels respectively.

Table 1 reveals a striking empirical pattern. A 10 percent increase in financial access is associated with an increase in consumption-income volatility of 0.09 among developing countries (column 1). In contrast, column 2 shows that a 10 percent improvement in financial access dampens consumption-income volatility by 0.15 in developed countries. Additionally, Panel B confirms the robustness of this result by excluding countries with extreme values of consumption-income volatility (> 3).

3 Model

We extend the small-open economy model of Aguiar and Gopinath (2007) (*AG* hereafter) by introducing an exogenous share of households without access to financial institutions.

3.1 Set-up

A representative firm produces the final good using labor (L_t) and capital (K_t) through a Cobb-Douglas production function:

$$Y_t = \exp(z_t)(\Gamma_t L_t)^{\alpha} K_t^{1-\alpha} \tag{2}$$

Here, α denotes the labor share; z_t captures transitory productivity; and Γ_t reflects trend productivity, evolving as follows:

$$\begin{aligned} z_t &= \rho_z z_{t-1} + \epsilon_t^z \; ; \; \epsilon_t^z \sim N\left(0, \sigma_z^2\right) & (3)\\ \Gamma_t &= \exp(g_t)\Gamma_{t-1} & (4)\\ g_t &= (1-\rho_g)\mu_g + \rho_g g_{t-1} + \epsilon_t^g \; ; \; \epsilon_t^g \sim N\left(0, \sigma_g^2\right) & (5) \end{aligned}$$

A trend shock (ϵ_t^g) permanently alters the future growth of output. The firm maximizes profits taking the wage rate (W_t) and rental rate (R_t) as given.

Households differ in financial access. A fraction $1-\lambda$ can save and borrow via a risk-free bond, while a fraction λ is financially excluded. These excluded households do not own any assets or hold any debt; they consume their income each period. An excluded household chooses consumption (C_t^e) and labor supply (L_t^e) to solve the following problem:

$$\begin{aligned} \max_{C_t^e, L_t^e} U\left(C_t^e, L_t^e\right) &= \frac{\left(C_t^e - \tau\Gamma_{t-1}\left(L_t^e\right)^{\eta}\right)^{1-\sigma}}{1-\sigma} \quad ; \tau > 0, \sigma > 0\; \eta > 1 & (6)\\ \text{subject to} &: \quad C_t^e = W_t L_t^e & (7) \end{aligned}$$

Including Γ_{t-1} in the utility function ensures stationarity of hours worked along the balanced growth path (Neumeyer & Perri, 2005).

Financially included households can purchase a risk-free bond at price q_t. They also accumulate physical capital to rent it to firms. However, capital depreciates at the rate δ and capital adjustment entails an adjustment cost (Mendoza, 1991). They choose consumption (C_t^a), labor (L_t^a), debt (B_{t+1}^a), and capital (K_{t+1}^a) every period to solve:

$$\max_{C_t^a, L_t^a, K_{t+1}^a, B_{t+1}^a} \quad \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \frac{\left(C_t^a - \tau \Gamma_{t-1} \left(L_t^a \right)^{\eta} \right)^{1-\sigma}}{1-\sigma} \right]; \ \beta \in (0,1) \tag{8}$$

$$\text{subject to:} \quad C_t^a + K_{t+1}^a - q_t B_{t+1}^a = W_t L_t^a + R_t K_t^a - B_t^a + (1-\delta) K_t^a - \frac{\phi}{2} K_t^a \left(\frac{K_{t+1}^a}{K_t^a} - \exp\left(\mu_g\right) \right)^2 \tag{9}$$

Following Schmitt-Grohé and Uribe (2003), bond pricing is determined internationally by:

$$\frac{1}{q_t} = R^* + \psi \left[\exp\left(\frac{B_{t+1}}{\Gamma_t} - b \right) - 1 \right] \tag{10}$$

where R^* is the global gross interest rate, b is the steady-state debt-to-GDP ratio, and ψ captures the elasticity of debt to interest rates.

Aggregate consumption (C_t) and labor supply (L_t) are weighted sums of consumption and hours worked of both household types:

$$C_t = \lambda C_t^e + (1-\lambda) C_t^a \tag{11}$$

$$L_t = \lambda L_t^e + (1-\lambda) L_t^a \tag{12}$$

3.2 Results

Following AG, we calibrate the model separately for Mexico (a developing economy) and Canada (a developed economy). To isolate the effect of financial access and technological shocks on business-cycle properties, we hold constant all parameters that affect steady-state outcomes.[2] We vary only the standard deviation of productivity shocks and the proportion of financially included households between Mexico and Canada. As we show below, this is sufficient for the model to replicate the key empirical patterns.

We adopt the following parameters from AG: β (discount factor) = 0.98; σ (risk-aversion) = 2;

[2]Our utility specification ensures that financial access heterogeneity does not influence steady-state consumption and output. See Appendices B.1 and B.2 for derivations.

α (labor share) = 0.68; δ (depreciation rate) = 5%; ϕ (capital adjustment cost) = 4; b (steady-state debt to GDP) = 10%, and, ψ (elasticity of debt to interest rate) = 0.1%. We set η = 1.6, yielding a Frisch elasticity $\left(\frac{1}{\eta-1}\right)$ of 1.7 (García-Cicco et al., 2010). The disutility of labor parameter, τ = 1.546, ensures steady-state labor equals one-third of available time. μ_g, trend growth rate of output, is set to $ln(1.007)$ based on the average trend growth rate of output in developing and developed economies.

We set λ based on the observed proportion of financial exclusion: 63.1% for Mexico and 0.03% for Canada. Table 2 shows that both income volatility and consumption-income volatility are higher in Mexico than in Canada. We estimate the standard deviation of trend (σ_g) and transitory (σ_z) productivity shocks by targeting these moments in the model simulations. For Mexico, we estimate $\sigma_g = 1.89$ and $\sigma_z = 0.01$; for Canada, $\sigma_g = 0.003$ and $\sigma_z = 0.5$.

Table 2: Model-fit: Business cycle moments for Mexico and Canada

	Mexico		Canada	
Business-cycle Moment	Data	Model	Data	Model
Income volatility, $\sigma(ln(Y))$	2.2	2.2	1.19	1.19
Consumption-Income volatility, $\sigma(ln(C))/\sigma(ln(Y))$	1.11	0.98	0.69	0.8

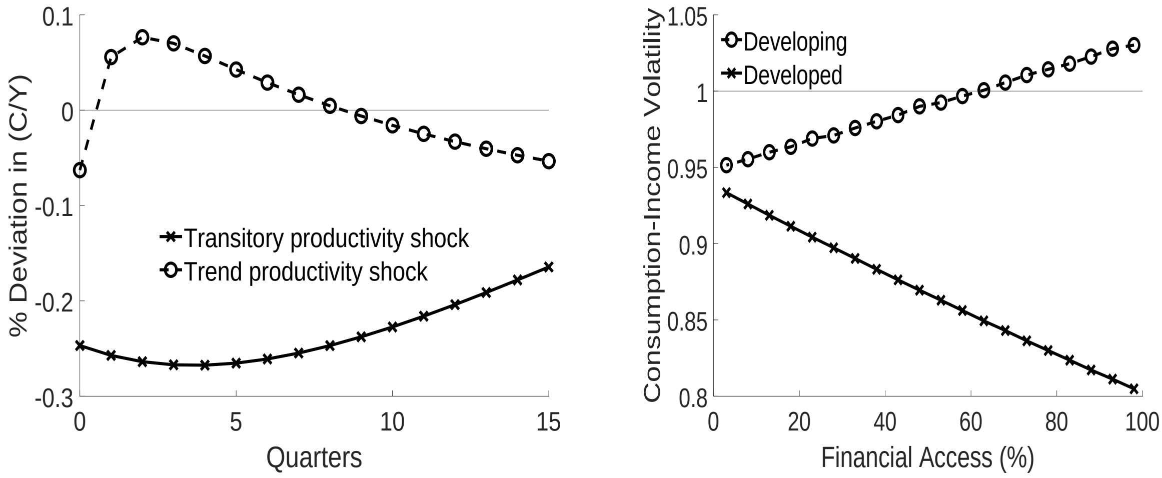
Note: The data moments for Mexico and Canada are computed over the period 1990-q1 to 2019-q4 using the IMF's International Financial Statistics. Model moments are obtained using simulations over 50,000 quarters.

The stark difference in the nature of productivity shocks between developing and developed countries explains our core finding. Using parameters of a developed economy, Panel (a) of Figure 1a illustrates that a positive transitory shock leads to a modest rise in consumption relative to income, as households save part of the temporary income gain. In contrast, a positive trend shock – which implies permanently higher expected future income – induces financially included households to borrow, raising consumption more than income. Consequently, aggregate consumption becomes significantly more volatile than output in response to trend compared to transitory shocks.

Panel (b) of Figure 1b highlights that greater financial inclusion increases consumption-income volatility in developing economies characterized by strong trend shocks. In contrast, since transitory shocks dominate in developed economies, higher financial access facilitates more consumption smoothing. Thus, the model reproduces the empirical pattern by varying only the variance of the productivity shocks.

Figure 1: Effect of financial access on consumption-income volatility in model

(a) Impulse Response of (C/Y) to productivity shocks | (b) Financial access and volatility



4 Discussion and Concluding Remarks

This study combines empirical evidence and theoretical analysis to examine the effect of financial access on consumption-income volatility in developing and developed countries. Using a comprehensive dataset covering 135 countries, we find that greater financial access increases consumption-income volatility in developing economies but reduces it in developed ones. This finding complements recent evidence based on narrower country samples (Bhattacharya & Patnaik, 2016; Donadelli & Gufler, 2024).

Building on recent work that shows greater financial access can amplify volatility through channels like interest rate shocks (Barrail, 2020) and labor market frictions (Epstein & Shapiro, 2021), our model highlights that the nature of technological shocks play a crucial role in mitigating or amplifying volatility for countries at different levels of development and financial access. Specifically, the response of permanent-income consumers to borrow (save) against trend (transitory) shocks significantly influences aggregate consumption dynamics – consistent with recent research (Guntin et al., 2023).

To isolate the role of technological shocks, we abstract from other frictions typically relevant in developing economies, such as financial frictions (Chang & Fernández, 2013; Hong, 2023) and commodity price shocks (Drechsel & Tenreyro, 2018). Future research can build on our framework by systematically decomposing the contributions of various shocks in explaining business-cycle dynamics of developing and developed economies using structural models.

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Appendix (Not for Publication)

A Data Sources and Descriptive Statistics

This section outlines the primary data sources, the construction of key variables, and the theoretical motivations for their inclusion in the regressions.

A.1 Data Sources and Sample Selection

Penn World Table

Data on national output, consumption, government expenditure, investment, exports and imports at constant 2017 prices and national population over the period 1970 to 2019 comes from the Penn World Table version 10.01 (Feenstra et al., 2015).

We measure trade-openness of a country as the ratio of the sum of exports and imports to GDP in 2019. Following Uribe and Schmitt-Grohé (2017), we include trade-openness to account for differences in volatility between more and less open-economies.

Hnatkovska and Loayza (2005) and Michaud and Rothert (2018) argue that developing countries undertake countercyclical government expenditure to insure against economic downturns – *fiscal impulse*. To build a measure of fiscal impulse, we begin by demeaning the ratio of HP-filtered government expenditure to GDP with the smallest absolute value for each country. We then compute the standard deviation of the growth of this ratio for each country.

We classify a country into one of five 5 regions: North America, Europe, Middle East and Centra Asia, Asia and Pacific, and Africa. We include region dummies to control for continent-specific factors.

Global Findex Database

Our measure of financial comes from the 2017 Global Findex Database (Demirgüç-Kunt & Klapper, 2012), which measures the proportion respondents who report having an account (by themselves or together with someone else) at a bank or another type of financial institution or using a mobile money service in the past year.

World Bank's World Development Indicators

Mendoza (1995) argue that terms-of-trade shocks can account for a large share of output variability. Data on terms-of-trade comes from the World Bank's World Development Indicators (The World Bank, 2024), where terms-of-trade measures the ratio of the export unit value index to the import unit value index. We include the standard deviation of a country's terms-of-trade between 1970-2019 in the regression.

We classify a country as developed if it falls under the World Bank's high-income category; otherwise, it is considered developing.

Flexible Exchange Rate Classification (Ilzetzki et al., 2021)

Levy-Yeyati and Sturzenegger (2003) argue that less flexible exchange rate regimes are linked to higher output volatility in developing countries . Following the coarse exchange rate classification and data from (Ilzetzki et al., 2021), we classify countries as operating under a flexible exchange rate system if their currency falls within a moving band narrower than $\pm$ 2%, or if it is categorized as managed floating, or freely floating, or freely falling.

Polity V Project

Existing research has shown that higher levels of democracy lead to higher growth and lower volatility (Acemoglu et al., 2019; Mobarak, 2005). We use data from the Polity V project (Marshall & Gurr, 2020), which constructs an index measuring the competitiveness of executive and political participation within a country. The index's value ranges from -10 to +10. We create a binary indicator of democracy if the index value is greater than 0 in 2019.

Banking, Currency and Debt Crisis

We use data provided by Laeven and Valencia (2020) on Banking, Currency and Debt crisis episodes to create a dummy whether the country experienced a Banking, Currency or Debt crisis between 1970 to 2017, respectively. They document that banking, currency and debt crises generate large output losses and macroeconomic volatility.

Sample Selection:

Following Uribe and Schmitt-Grohé (2017), countries with less than 20 years of uninterrupted output, consumption, investment, government expenditure and net exports data are dropped. We also drop countries with less than 1 million population in 2019. Additionally, we exclude United States (US) from the sample, as the model we present better captures features of a small open-economy rather than a large open-economy like the US. We drop Iran and Iraq due to extremely large fluctuations in output and consumption. We drop Taiwan because of unavailability of terms-of-trade data.

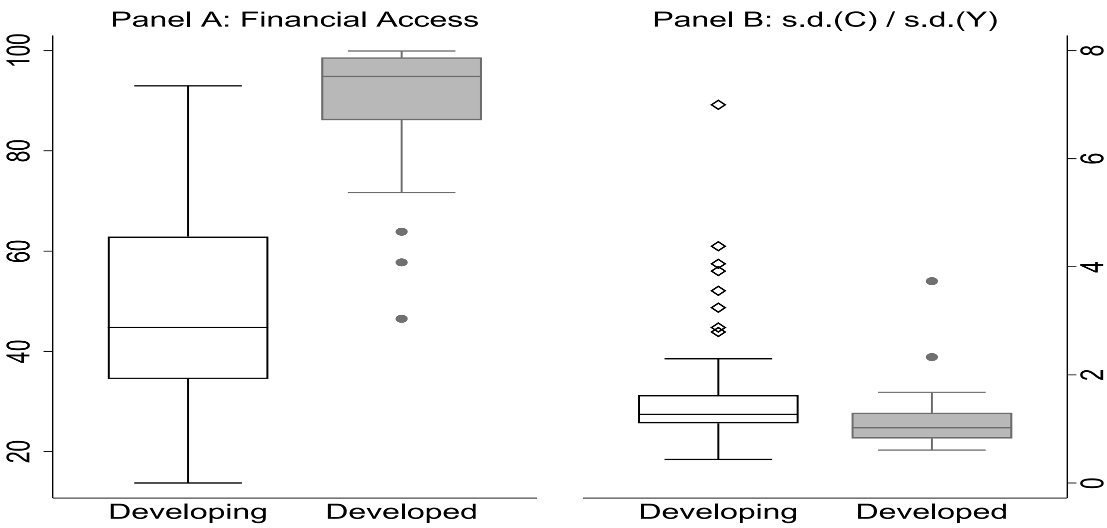
Table A.1: List of Countries in sample by World Bank Classification

Developed	Developing		
	Upper-Middle Income	Lower-Middle Income	Low Income
Australia	Albania	Algeria	Afghanistan
Austria	Argentina	Bangladesh	Burkina Faso
Bahrain	Armenia	Benin	Central African Republic
Belgium	Azerbaijan	Bolivia	Chad
Canada	Belarus	Cambodia	D.R. Congo
Chile	Bosnia & Herzegovina	Cameroon	Ethiopia
Croatia	Botswana	Congo	Gambia
Czech Republic	Brazil	Egypt	Guinea
Denmark	Bulgaria	El Salvador	Haiti
Estonia	China	Ghana	Liberia
Finland	Colombia	Honduras	Madagascar
France	Costa Rica	India	Malawi
Germany	Dominican Republic	Ivory Coast	Mali
Greece	Ecuador	Kenya	Mozambique
Hong Kong SARC	Gabon	Kyrgyzstan	Niger
Hungary	Georgia	Lao People's DR	Rwanda
Ireland	Guatemala	Lesotho	Sierra Leone
Israel	Indonesia	Mauritania	Tajikistan
Italy	Jordan	Mongolia	Togo
Japan	Kazakhstan	Morocco	Uganda
Kuwait	Lebanon	Myanmar	
Latvia	Libya	Nepal	
Lithuania	Malaysia	Nicaragua	
Mauritius	Mexico	Nigeria	
Netherlands	Namibia	Pakistan	
New Zealand	Paraguay	Philippines	
Norway	Peru	Rep. of Moldova	
Panama	Russian	Senegal	
Poland	Serbia	Sri Lanka	
Portugal	South Africa	State of Palestine	
Republic of Korea	TFYR of Macedonia	Tunisia	
Romania	Thailand	U.R. of Tanzania	
Saudi Arabia	Turkey	Ukraine	
Singapore	Turkmenistan	Uzbekistan	
Slovakia	Venezuela	Viet Nam	
Slovenia		Zambia	
Spain		Zimbabwe	
Sweden			
Switzerland			
Trinidad and Tobago			
United Arab Emirates			
United Kingdom			
Uruguay			

A.2 Descriptive Statistics

Figure A.1 shows a box-whisker plot of financial access and consumption-income volatility for developed and developing economies in Panel A and B, respectively. Panel A shows that the median proportion of households with financial access is much higher in developed (95%) than

Figure A.1: Box-whisker plot of financial access and consumption-income volatility



Note: The top and bottom range shows the 1.5 $\times$ inter-quartile range. All statistics are weighted by the log population in 2019 of each country.

developing countries (45%). Panel B highlights that the median consumption-income volatility is 1.27 and 1.02 in developing and developed economies, respectively.

B Model FOCs, Solution and Calibration

We derive the first-order conditions (F.O.C.s) and state the definition of competitive equilibrium in Section B.1. Since, the model features non-stationary productivity process, we discuss de-trending of the model and steady-state of de-trended model in Section B.2.

B.1 Model Derivation

B.1.1 First-order conditions

The representative firm maximizes profits:

$$\max_{L_t, K_t} Y_t - W_t L_t - R_t K_t \tag{B.1}$$

where, the price of the numéraire good is 1. The F.O.C.s with respect to labor and capital are:

$$\alpha Y_t = W_t L_t \tag{B.2}$$

$$(1-\alpha)\, Y_t = R_t K_t \tag{B.3}$$

The first-order condition w.r.t. labor supply for a financially excluded household is:

$$W_t = \eta \tau \Gamma_{t-1} \left(L_t^e\right)^{\eta-1} \tag{B.4}$$

The F.O.C.s with respect to consumption, labor supply, capital and bond holding for the households with financial access are:

$$\begin{aligned}
\beta^t \left(C_t^a - \tau \Gamma_{t-1} \left(L_t^a\right)^{\eta}\right)^{-\sigma} &= \gamma_t && \text{(B.5)}\\
\beta^t \eta \tau \Gamma_{t-1} \left(C_t^a - \tau \Gamma_{t-1} \left(L_t^a\right)^{\eta}\right)^{-\sigma} \left(L_t^a\right)^{\eta-1} &= \gamma_t W_t && \text{(B.6)}\\
q_t \gamma_t &= \mathrm{E}_t \gamma_{t+1} && \text{(B.7)}\\
\mathrm{E}_t \gamma_{t+1} \left[1 - \delta + R_{t+1} + \frac{\phi}{2} \left\{ \left(\frac{K_{t+1}^a}{K_t^a}\right)^2 - exp\left(2\mu_g\right) \right\}\right] &= \left[1 + \phi \left(\frac{K_{t+2}^a}{K_{t+1}^a} - exp\left(\mu_g\right)\right)\right] \gamma_t && \text{(B.8)}
\end{aligned}$$

where, γ_t is the Lagrange multiplier corresponding to the per-period budget constraint.

The labor supply equation is then given by combining equations B.5 and B.6 :

$$W_t = \eta \tau \Gamma_{t-1} \left(L_t^a\right)^{\eta-1} \tag{B.9}$$

B.1.2 Aggregation and Equilibrium

The labor supply equations of the financially included (equation B.9) and excluded households (equation B.4) imply that

$$L_t^a = L_t^e = L_t = \left(\frac{W_t}{\eta\tau\Gamma_{t-1}}\right)^{\frac{1}{\eta-1}} \tag{B.10}$$

Only financially included households supply capital and demand bonds, which implies the following relationships for aggregate debt (D_t) and aggregate capital stock (K_t):

$$K_t = (1-\lambda)K_t^a \tag{B.11}$$

$$B_t = (1-\lambda)B_t^a \tag{B.12}$$

To derive the the Euler equation for aggregate investment, we first re-write the Lagrange multiplier (equation B.5) by substituting $C_t^a = \frac{C_t - \lambda C_t^e}{1-\lambda}$ and $L_t^e = L_t$:

$$\beta^t\left(\frac{C_t - \lambda C_t^e}{1-\lambda} - \tau\Gamma_{t-1}\left(L_t\right)^{\eta}\right)^{-\sigma} = \gamma_t \tag{B.13}$$

$$\beta^t\left(\frac{C_t - \tau\Omega\Gamma_{t-1}\left(L_t\right)^{\eta}}{1-\lambda}\right)^{-\sigma} = \gamma_t \tag{B.14}$$

where, the last equality uses the budget constraint of the financially excluded households wherein $C_t^e = \eta\tau\Gamma_{t-1}\left(L_t^e\right)^{\eta}$ and $\Omega = 1-\lambda+\lambda\eta$. Thus, the Euler equation for the aggregate capital demand is:

$$\left[1+\phi\left(\frac{K_{t+1}^a}{K_t^a} - exp\left(\mu_g\right)\right)\right] \times \left(C_t - \tau\Omega\Gamma_{t-1}\left(L_t\right)^{\eta}\right)^{-\sigma} =$$
$$\beta \mathrm{E}_t\left[\left(C_{t+1} - \tau\Omega\Gamma_t\left(L_{t+1}\right)^{\eta}\right)^{-\sigma} \times \left[1-\delta+R_{t+1}+\frac{\phi}{2}\left\{\left(\frac{K_{t+2}^a}{K_{t+1}^a}\right)^2 - exp\left(2\mu_g\right)\right\}\right]\right] \tag{B.15}$$

Similarly, the Euler equation for the aggregate level of debt is:

$$\left(C_t - \tau\Omega\Gamma_{t-1}\left(L_t\right)^{\eta}\right)^{-\sigma} q_t = \beta \mathrm{E}_t\left[\left(C_{t+1} - \tau\Omega\Gamma_t\left(L_{t+1}\right)^{\eta}\right)^{-\sigma}\right] \tag{B.16}$$

Recall the budget constraint of the financially included households:

$$C_t^a + K_{t+1}^a - q_t B_{t+1}^a = W_t L_t^a + R_t K_t^a - B_t^a + (1-\delta)K_t^a - \frac{\phi}{2}K_t^a\left(\frac{K_{t+1}^a}{K_t^a} - exp\left(\mu_g\right)\right)^2 \tag{B.17}$$

To derive the aggregate resource constraint, we substitute $C_t^a = \frac{C_t - \lambda C_t^e}{1-\lambda}$, $L_t^e = \frac{L_t - \lambda L_t^e}{1-\lambda}$, $K_t^a = \frac{K_t}{1-\lambda}$ and $B_t^a = \frac{B_t}{1-\lambda}$, which gives us:

$$
\begin{aligned}
\frac{C_t - \lambda C_t^e}{1-\lambda} + \frac{K_{t+1}}{1-\lambda} - q_t \frac{B_{t+1}}{1-\lambda} &= W_t \frac{L_t - \lambda L_t^e}{1-\lambda} + R_t \frac{K_t}{1-\lambda} - \frac{B_t}{1-\lambda} + (1-\delta)\frac{K_t}{1-\lambda} - \frac{\phi}{2}\frac{K_t}{1-\lambda}\left(\frac{K_{t+1}}{K_t} - exp\left(\mu_g\right)\right)^2 \\
\left(C_t - \lambda C_t^e\right) + K_{t+1} - q_t B_{t+1} &= W_t\left(L_t - \lambda L_t^e\right) + R_t K_t - B_t + (1-\delta)K_t - \frac{\phi}{2}K_t\left(\frac{K_{t+1}}{K_t} - exp\left(\mu_g\right)\right)^2
\end{aligned}
$$

Since, $C_t^e = W_t L_t^e$, this implies:

$$
C_t + K_{t+1} - q_t B_{t+1} = W_t L_t + R_t K_t - B_t + (1-\delta)K_t - \frac{\phi}{2}K_t\left(\frac{K_{t+1}}{K_t} - exp\left(\mu_g\right)\right)^2
$$

The representative firm's first-order conditions (equations B.2 and B.3) jointly imply that $Y_t = W_t L_t + R_t K_t$. Hence, the aggregate resource constraint is:

$$
C_t + I_t + NX_t = Y_t \tag{B.18}
$$

where, net exports NX_t and Investment I_t are defined as:

$$
\begin{aligned}
NX_t &= B_t - q_t B_{t+1} && \text{(B.19)} \\
I_t &= K_{t+1} - (1-\delta)K_t + \frac{\phi}{2}K_t\left(\frac{K_{t+1}}{K_t} - exp\left(\mu_g\right)\right)^2 && \text{(B.20)}
\end{aligned}
$$

In equilibrium, households with and without access to the financial markets and the representative firm solve their respective optimization problem given the initial capital stock K_{-1}, the initial level of debt B_{-1}, the initial labor-augmenting productivity Γ_{-1} and the sequences of debt prices $\{q_t\}_{t=0}^{\infty}$, transitory productivity shocks $\{z_t\}_{t=0}^{\infty}$, growth process $\{\Gamma_t\}_{t=0}^{\infty}$ and trend shocks $\{g_t\}_{t=0}^{\infty}$. A competitive equilibrium is a set of quantities $\{C_t, L_t, K_{t+1}, B_{t+1}, Y_t, I_t, NX_t\}_{t=0}^{\infty}$ and prices $\{W_t, R_t\}_{t=0}^{\infty}$ such that:

1. The representative firm chooses L_t, K_t using the first-order conditions (equations B.2 and B.3) given the production technology (equations 2, 3, 4 and 5)

2. Households choose consumption and labor supply given their budget constraint and price of debt (equation 10) such that the aggregate labor supply (equation B.10), and Euler equations for aggregate investment and debt (equations B.15 and B.16) hold.

3. Aggregate resource constraint (equation B.18) holds given the definition of net exports

(equation B.19) and capital investment (equation B.20).

B.2 Model Solution

We de-trend variables using Γ_{t-1}, as total factor productivity is non-stationary with a stochastic trend. Let $\tilde{x}_t$ denote x_t's de-trended counterpart. A competitive equilibrium then is a set of allocations $\{\tilde{C}_t, L_t, \tilde{K}_{t+1}, \tilde{B}_{t+1}, \tilde{Y}_t, \tilde{I}_t, \tilde{NX}_t\}_{t=0}^{\infty}$ and factor prices $\{\tilde{W}_t, R_t\}_{t=0}^{\infty}$ given sequences of productivity shocks $\{z_t, g_t\}_{t=0}^{\infty}$ and debt prices $\{q_t\}_{t=0}^{\infty}$ satisfying:

1. The production function and productivity shocks :

$$\tilde{Y}_t = exp(z_t)\, exp(\alpha g_t)\, L_t^{\alpha} \tilde{K}_t^{1-\alpha} \tag{B.21}$$

$$z_t = \rho_z z_{t-1} + \epsilon_t^z \tag{B.22}$$

$$g_t = (1-\rho_g)\mu_g + \rho_g g_{t-1} + \epsilon_t^g \tag{B.23}$$

2. labor demand and capital demand equations

$$\alpha \tilde{Y}_t = \tilde{W}_t L_t \tag{B.24}$$

$$(1-\alpha)\, \tilde{Y}_t = R_t \tilde{K}_t \tag{B.25}$$

3. Household labor supply equation:

$$L_t = \left(\frac{\tilde{W}_t}{\eta\tau}\right)^{\frac{1}{\eta-1}} \tag{B.26}$$

4. Euler equation of aggregate bond holdings:

$$\left(\tilde{C}_t - \tau\Omega\,(L_t)^{\eta}\right)^{-\sigma} q_t = \beta exp(-\sigma g_t)\, \mathrm{E}_t\left[\left(\tilde{C}_{t+1} - \tau\Omega\,(L_{t+1})^{\eta}\right)^{-\sigma}\right] \tag{B.27}$$

5. Euler equation of aggregate investment:

$$\left(\tilde{C}_t - \tau\Omega\,(L_t)^{\eta}\right)^{-\sigma} \times \left[1 + \phi\left(\frac{\tilde{K}_{t+1}^a}{\tilde{K}_t^a} exp(g_t) - exp(\mu_g)\right)\right] = \beta exp(-\sigma g_t) \times$$
$$\mathrm{E}_t\left[\left(\tilde{C}_{t+1} - \tau\Omega\,(L_{t+1})^{\eta}\right)^{-\sigma} \times \left[1 - \delta + R_{t+1} + \frac{\phi}{2}\left\{\left(\frac{\tilde{K}_{t+2}^a}{\tilde{K}_{t+1}^a} exp(g_{t+1})\right)^2 - exp(2\mu_g)\right\}\right]\right] \tag{B.28}$$

6. Price of debt:

$$\frac{1}{q_t} = R^* + \psi\left[exp\left(\tilde{B}_{t+1} - b\right) - 1\right] \tag{B.29}$$

7. Aggregate resource constraint:

$$\begin{aligned}
\tilde{Y}_t &= \tilde{C}_t + \tilde{I}_t + \tilde{NX}_t && \text{(B.30)}\\
\tilde{NX}_t &= \tilde{B}_t - q_t exp\left(g_t\right)\tilde{B}_{t+1} && \text{(B.31)}\\
\tilde{I}_t &= \tilde{K}_{t+1} exp\left(g_t\right) - (1-\delta)\tilde{K}_t + \frac{\phi}{2}\tilde{K}_t\left(\frac{\tilde{K}_{t+1}}{\tilde{K}_t} exp\left(g_t\right) - exp\left(\mu_g\right)\right)^2 && \text{(B.32)}
\end{aligned}$$

B.2.1 Steady-state

Let $\bar{x}$ denote the steady-state of variable x_t. Below, we derive the steady-state values of aggregate variables and prices in the model. The key point to note is that they will be independent of household heterogeneity. The steady-state of the technological shocks z_t and g_t are 0 and μ_g, respectively. The Euler equations provide the steady-state values of q_t and R_t:

$$\begin{aligned}
\bar{q} &= \beta exp\left(-\sigma\mu_g\right) && \text{(B.33)}\\
\bar{R} &= \frac{1}{\bar{q}} - (1-\delta) && \text{(B.34)}
\end{aligned}$$

The firm's labor and capital demand equations imply that in steady-state:

$$\bar{\tilde{W}} = \frac{\alpha}{1-\alpha}\bar{R}\frac{\bar{\tilde{K}}}{L} \tag{B.35}$$

Manipulating the capital demand equation given $\bar{R}$ and the production function gives $\frac{\bar{\tilde{K}}}{L} = exp\left(\mu_g\right) / \left(\frac{\bar{R}}{(1-\alpha)}\right)^{1/\alpha}$. Thus, steady-state de-trended wage rate is:

$$\bar{\tilde{W}} = exp\left(\mu_g\right)\alpha\left(\frac{\bar{R}}{(1-\alpha)}\right)^{\frac{\alpha-1}{\alpha}} \tag{B.36}$$

The steady-state values of labor, de-trended capital and de-trended output are:

$$\bar{L} = \left(\frac{\bar{\tilde{W}}}{\eta\tau}\right)^{\frac{1}{\eta-1}} \tag{B.37}$$

$$
\begin{aligned}
\bar{\tilde{K}} &= \frac{\bar{\tilde{W}}\bar{L}}{\bar{R}}\frac{1-\alpha}{\alpha} && \text{(B.38)}\\
\bar{\tilde{Y}} &= exp\left(\alpha\mu_g\right)\bar{L}^{\alpha}\bar{\tilde{K}}^{1-\alpha} && \text{(B.39)}
\end{aligned}
$$

Following Aguiar and Gopinath (2007), assume that the steady-state debt to GDP ratio is b. The steady-state values of de-trended debt, net exports, investment and consumption are:

$$
\begin{aligned}
\bar{\tilde{B}} &= b \times \bar{\tilde{Y}} && \text{(B.40)}\\
\bar{\tilde{NX}} &= \bar{\tilde{B}}\left(1-\bar{q}exp\left(\alpha\mu_g\right)\right) && \text{(B.41)}\\
\bar{\tilde{I}} &= \bar{\tilde{K}}\left(exp\left(\alpha\mu_g\right)+\delta-1\right) && \text{(B.42)}\\
\bar{\tilde{C}} &= \bar{\tilde{Y}}-\bar{\tilde{I}}-\bar{\tilde{NX}} && \text{(B.43)}
\end{aligned}
$$

Note that the utility function formulation implies that aggregate consumption is independent of hours worked and hence, aggregate consumption and output in steady-state are independent of the share of households without financial access. Thus, the steady-state outcomes of developed and developing economies are equal.

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